

Graph Generation through Self-alignment in Transformer-based Discrete Diffusion Model

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Abstract—Discrete diffusion models provide a principled framework for graph generation. Despite their success, existing discrete graph diffusion models primarily rely on output-level denoising objectives, leaving intermediate representations largely unconstrained. Recent advances in diffusion transformers suggest that representation alignment or self-alignment can serve as an effective mechanism for regularizing intermediate representations without altering the underlying architecture. However, such ideas have not been systematically studied in the context of discrete graph diffusion. In this work, we bridge this gap by incorporating self-alignment into discrete graph diffusion models. In our SAGraphDiff model, we align intermediate transformer-based graph representations at the same diffusion step through a lightweight auxiliary objective with negligible computational overhead while fully preserving the original diffusion formulation. Experiments on molecular graph benchmarks demonstrate that self-alignment improves the reliability of one-shot graph generation, leading to higher molecular validity and enhanced novelty while maintaining competitive performance on other metrics.

Index Terms—graph generation, diffusion models, representation alignment

I. INTRODUCTION

Diffusion models [1] have recently emerged as a powerful paradigm for generative modeling, achieving strong empirical performance across a variety of domains. Motivated by these advances, there has been growing interest in extending diffusion-based approaches to graph-structured data, particularly graph generation [2], which is central to many scientific applications.

Unlike continuous domains such as images, graphs are discrete and combinatorial objects whose node indices are arbitrary and carry no semantic meaning. As a result, graph generative models must satisfy permutation invariance at the distribution level, while relying on permutation-equivariant transformations to process node- and edge-wise representations [3]. These requirements, together with the presence of complex structural constraints, make the direct application of continuous Gaussian diffusion non-trivial [4]. Discrete diffusion models [5], [6] address these challenges by defining forward and reverse processes directly in the discrete graph domain. In particular, one-shot discrete graph diffusion [4], which generates all nodes and edges simultaneously in a single denoising pass, avoids imposing an explicit node ordering

and enables permutation-consistent generation. This stands in contrast to those earlier sequential graph generation methods [3], [7], which often rely on arbitrary orderings and may suffer from ordering bias [3].

Despite their success, existing discrete graph diffusion models primarily focus on output-level denoising objectives, aiming to reconstruct clean graphs from noisy observations [4]. Recent advances in diffusion models have shown that incorporating representation-level objectives, such as representation alignment or self-alignment, can improve training efficiency and generative performance [8].

In this work, we introduce **self-alignment** as a general auxiliary training strategy for discrete graph diffusion models. Specifically, we propose **SAGraphDiff**, which builds graph representation self-alignment on top of DiGress [4], a representative discrete diffusion framework for one-shot graph generation. Our contributions can be summarized as follows:

- We present the first study that introduces self-alignment into discrete diffusion models for graph generation. Our work highlights a representation-level perspective in discrete graph diffusion that has not yet been systematically explored, extending representation alignment from continuous diffusion models to discrete, one-shot graph generation.
- We instantiate graph representation self-alignment on DiGress and develop a practical training framework for incorporating representation-level regularization into one-shot discrete graph diffusion models. In our setting, self-alignment is applied as a lightweight auxiliary objective during training, without modifying the underlying diffusion formulation or model architecture, making it easy to integrate into existing discrete graph diffusion pipelines.
- We conduct extensive experiments on molecular graph generation benchmarks and observe that incorporating self-alignment leads to overall improvements in generation performance compared to the baseline. Self-alignment reduces the training time required to reach strong generation performance, highlighting its effectiveness as an auxiliary regularization for discrete graph diffusion.

*Equal contribution.

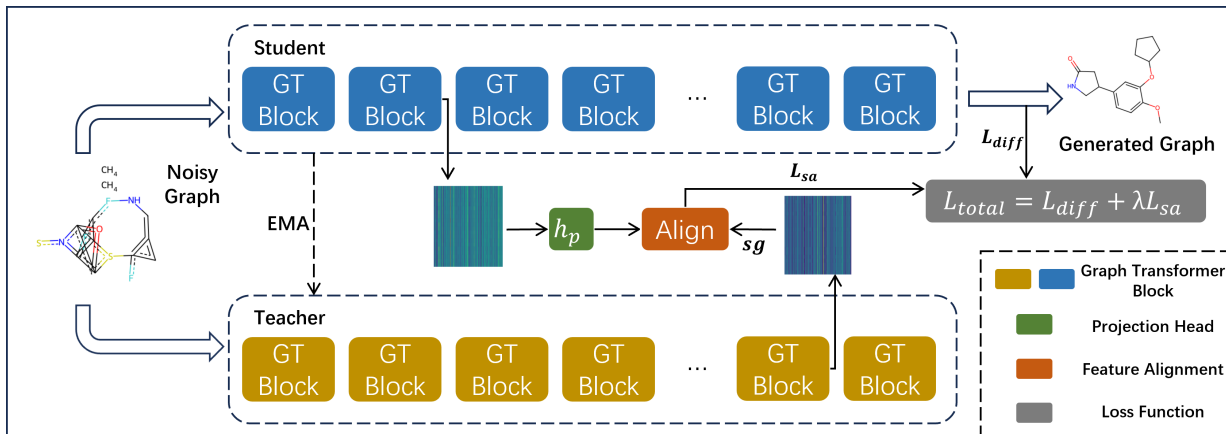


Fig. 1. **Overall framework of SAGraphDiff.** Given a noisy graph G_t , a student model predicts the reverse diffusion process and is trained with the diffusion loss $\mathcal{L}_{diff}$. To enhance representation consistency, self-alignment aligns the student’s intermediate representation from an earlier layer (under higher noise) with the teacher’s representation from a later layer (under lower noise) via a projection head h_p and an alignment objective $\mathcal{L}_{sa}$. The teacher is updated as an exponential moving average (EMA) of the student, and a stop-gradient operator (sg) is applied to the teacher to prevent gradient flow. The total training objective is $\mathcal{L}_{total} = \mathcal{L}_{diff} + \lambda \mathcal{L}_{sa}$.

II. RELATED WORK

A. Graph Generation

Graph generation [7] has been extensively studied due to its broad applications in molecular design, social networks [3], and knowledge graphs. More recently, deep generative models have become the dominant paradigm, aiming to learn graph distributions directly from data [7]. Representative approaches include autoregressive models [3] that sequentially add nodes and edges, variational models [9] that optimize an evidence lower bound with latent representations, and adversarial methods [10] that match generated and real graph distributions.

While these methods have achieved notable success, they often introduce ordering bias or rely on continuous relaxations, complicating the handling of discrete molecular graphs where chemical validity and structural constraints must be jointly satisfied [4].

B. Discrete Diffusion Models for Graph Generation

Diffusion models [1] have shown strong generative performance, but directly applying continuous Gaussian diffusion to graphs is non-trivial due to the mismatch between additive Gaussian noise and discrete graph states [4], [5]. Discrete diffusion models [4], [6] address this by defining forward noising processes over discrete state spaces and learning a reverse denoising process. A representative model is DiGress [4], which performs one-shot graph generation directly in the discrete domain without relying on explicit node orderings, naturally preserving permutation consistency. However, existing discrete graph diffusion models optimize only output-level denoising objectives, leaving intermediate representations implicitly constrained, which motivates our work.

C. Representation Alignment in Diffusion Models

A growing body of work studies representation alignment, self-distillation, and auxiliary representation-level objectives to improve diffusion model training and sampling. Many of

these methods [8] were developed for continuous diffusion models, particularly in image generation, where intermediate features can be encouraged to be more consistent, informative, or semantically aligned. Existing approaches can be broadly categorized into (i) external alignment methods that match diffusion representations to pretrained encoders (e.g., CLIP- or self-supervised backbones) [11], and (ii) self-alignment / self-distillation strategies that impose consistency across timesteps, network layers, or teacher–student snapshots within the diffusion model itself [8]. These methods typically assume continuous and dense latent spaces (pixel/latent diffusion), where similarity objectives (e.g., cosine, contrastive, feature regression) are naturally defined. Discrete graph diffusion, however, presents a unique challenge: while graph states are categorical and lack a natural similarity metric, their intermediate transformer representations remain continuous, making representation-space alignment feasible without modifying the diffusion formulation. Despite this opportunity, the applicability of such techniques to discrete graph diffusion has not been systematically studied, which our work aims to address.

III. METHOD

A. Overview

As shown in Figure 1, we propose SAGraphDiff, which augments a one-shot discrete graph diffusion model with a lightweight alignment objective on intermediate representations. We instantiate our approach on DiGress [4], a canonical backbone for discrete graph diffusion, which allows us to isolate the effect of representation alignment without introducing confounding architectural changes.

B. Preliminaries: Discrete Graph Diffusion

a) *Graph representation.*: We consider graphs with discrete node and edge attributes. Let n denote the number of nodes. Node attributes are represented as $\mathbf{X} \in \{1, \dots, C_x\}^n$, and edge attributes as $\mathbf{E} \in \{1, \dots, C_e\}^{n \times n}$. The clean graph

is denoted by $G_0 = (\mathbf{X}_0, \mathbf{E}_0)$, and its corrupted version at diffusion step t is $G_t = (\mathbf{X}_t, \mathbf{E}_t)$.

b) Forward discrete diffusion.: Discrete diffusion defines a Markov noising process over categorical variables:

$$q(G_{1:T} | G_0) = \prod_{t=1}^T q(G_t | G_{t-1}), \quad (1)$$

where transitions factorize over nodes and edges:

$$q(G_t | G_{t-1}) = \prod_{i=1}^n q(x_t^{(i)} | x_{t-1}^{(i)}) \prod_{i,j=1}^n q(e_t^{(ij)} | e_{t-1}^{(ij)}). \quad (2)$$

Each transition is parameterized by a time-dependent categorical transition matrix, which gradually increases uncertainty by replacing clean symbols with noise or masked states. This formulation operates directly in the discrete domain and preserves permutation invariance with respect to node indexing.

c) Reverse denoising process.: The reverse process is modeled by a neural denoiser p_θ that predicts clean discrete states from noisy graphs:

$$p_\theta(G_{t-1} | G_t) = \prod_{i=1}^n p_\theta(x_{t-1}^{(i)} | G_t, t) \prod_{i,j=1}^n p_\theta(e_{t-1}^{(ij)} | G_t, t). \quad (3)$$

In practice, the model outputs categorical distributions over node and edge attributes, parameterized by permutation-invariant graph neural networks.

d) Training objective.: Training is performed by minimizing a categorical denoising objective, which can be written as a sum of cross-entropy losses:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t, G_0, G_t} \left[\mathcal{L}_{\text{CE}}(\mathbf{X}_0, \hat{\mathbf{X}}_\theta(G_t, t)) + \mathcal{L}_{\text{CE}}(\mathbf{E}_0, \hat{\mathbf{E}}_\theta(G_t, t)) \right]. \quad (4)$$

This objective supervises only the final prediction of discrete graph states, while intermediate representations are optimized implicitly.

C. Representation Extraction in Graph Diffusion

During the reverse denoising process, the diffusion model produces intermediate representations at multiple network layers and noise levels. In our setting, the denoising network is implemented using a Transformer-based architecture, which naturally yields a hierarchy of intermediate features across layers. These representations encode partial structural and semantic information about the graph as denoising progresses. Earlier layers or representations associated with higher noise levels tend to be weaker and more ambiguous, while deeper layers or lower-noise representations capture stronger and more stable graph structure [12]. As illustrated in Figure 2, the inter-layer cosine similarity matrix computed from a trained DiGress model reveals that adjacent layers exhibit high representational similarity (e.g., $\text{sim}(\text{L3}, \text{L4}) = 0.899$), indicating that discrete diffusion training tends to produce homogeneous intermediate representations. At the same time, layers with greater depth separation retain meaningful differences (e.g.,

Algorithm 1 Training Algorithm for SAGraphDiff

Input: A graph $G_0 = (X_0, E_0)$; student model θ_s ; teacher model θ_t (EMA); projection head h_p ; graph self-alignment weight λ_{sa}

Sample $t \sim \mathcal{U}(1, \dots, T)$

Sample $G_t \sim q(G_t | G_0)$ $\triangleright$ Sample a (discrete) noisy graph

$z \leftarrow f(G_t, t)$ $\triangleright$ Structural and spectral features

$\hat{p}^X, \hat{p}^E \leftarrow p_{\theta_s}(G_t, t)$ $\triangleright$ Forward pass

$\mathcal{L}_{\text{diff}} \leftarrow \mathcal{L}_{\text{CE}}(\hat{p}^X, X_0) + \lambda \mathcal{L}_{\text{CE}}(\hat{p}^E, E_0)$ $\triangleright$ Cross-entropy

if t is in early diffusion steps **then**

$H_s \leftarrow \text{Rep}_{\theta_s}^{(t_s)}(G_t, t)$ $\triangleright$ Earlier-layer student representation (high noise)

$t' \leftarrow \max(t - \delta_t, 1), \quad \delta_t \sim \mathcal{U}(0, [0.1T])$ $\triangleright$ Sample lower-noise teacher timestep

$G_{t'} \sim q(G_{t'} | G_0)$ $\triangleright$ Generate teacher noisy graph

$H_t \leftarrow \text{Rep}_{\theta_t}^{(t')}(G_{t'}, t')$ $\triangleright$ Later-layer teacher (EMA) representation (low noise)

$z_s \leftarrow \text{Pool}(H_s), z_t \leftarrow \text{Pool}(H_t)$ $\triangleright$ Graph-level pooling

$z'_s \leftarrow h_p(z_s)$ $\triangleright$ Projection head

$\mathcal{L}_{\text{sa}} \leftarrow \text{SmoothL1}(z'_s, \text{stopgrad}(z_t))$ $\triangleright$ sa loss

$\mathcal{L} \leftarrow \mathcal{L}_{\text{diff}} + \lambda_{\text{sa}} \mathcal{L}_{\text{sa}}$

else

$\mathcal{L} \leftarrow \mathcal{L}_{\text{diff}}$

end

optimizer.step($\mathcal{L}$)

$\theta_t \leftarrow \alpha \theta_t + (1 - \alpha) \theta_s$ $\triangleright$ EMA update

$\text{sim}(\text{L3}, \text{L8}) = 0.546$). This motivates aligning shallow student representations toward deeper teacher representations as an explicit cross-layer supervisory signal. These representations

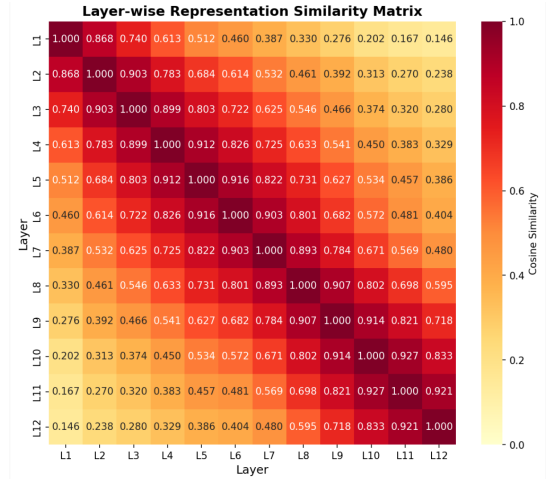


Fig. 2. Inter-layer cosine similarity matrix of a trained DiGress model. Adjacent layers show high similarity, while layers with greater depth separation retain meaningful representational differences, motivating cross-layer alignment.

are extracted from the standard forward pass without additional encoders, and a lightweight projection head [8] is applied for computing the alignment objective.

D. Self-Alignment of Graph Representation

a) Student and teacher representations.: Let student representation $H_s \leftarrow \text{Rep}_{\theta_s}^{(t_s)}(G_t, t)$ denote an intermediate

representation extracted from layer l_s at diffusion step t . Student representations are selected from earlier layers of the denoising network, where representations tend to be weaker and less stable. Correspondingly, teacher representations are obtained as $H_t \leftarrow \text{Rep}_{\theta_t}^{(l_t)}(G_{t'}, t')$, where $l_t > l_s$. The teacher parameters θ_t are updated as an exponential moving average (EMA) of the student parameters θ_s :

$$\theta_t \leftarrow \alpha \theta_t + (1 - \alpha) \theta_s \quad (5)$$

EMA averaging produces a temporally smoothed parameter trajectory, yielding more stable and consistent representations than the instantaneous student parameters [13]. This stability makes EMA teacher representations particularly suitable as alignment targets, as they are less susceptible to training noise and provide reliable guidance throughout optimization. This cross-layer supervision encourages weaker representations to learn from the more informative outputs of deeper layers, providing an additional training signal where the denoising objective alone offers limited guidance. Following [8], we apply representation alignment only at early diffusion steps satisfying $t/T \leq 0.2$, and disable it for later stages. This design is motivated by the observation that in discrete diffusion, high-noise steps correspond to the most ambiguous denoising conditions, where intermediate representations are least constrained by the denoising objective and can benefit most from auxiliary regularization [4]. In contrast, representations at later diffusion steps are already strongly constrained by the denoising objective and closely tied to the prediction of clean graph variables, making additional alignment objectives less beneficial and potentially disruptive [13].

b) *Graph self-alignment loss.*: Given a student representation H_s and its corresponding teacher representation H_t , where H_s is computed at diffusion step t and H_t is computed at a lower-noise step t' , we define the graph self-alignment loss as

$$\mathcal{L}_{\text{sa}} = \text{SmoothL1}\left(h_p(\text{Pool}(H_s)), \text{stopgrad}(\text{Pool}(H_t))\right). \quad (6)$$

where $h_p(\cdot)$ denotes a projection head, and $\text{Pool}(\cdot)$ denotes the Mean-Pooling Function. The SmoothL1 loss is defined as:

$$\text{SmoothL1}(x) = \begin{cases} \frac{1}{2\beta} x^2, & \text{if } |x| < \beta \\ |x| - \frac{\beta}{2}, & \text{otherwise} \end{cases} \quad (7)$$

where $\beta = 0.05$ is the threshold parameter controlling the transition between the quadratic and linear regimes [8]. Gradients are not propagated through the teacher branch.

E. Overall Training Objective & Procedure

The final training objective combines the standard diffusion loss with the graph self-alignment regularization:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \mathcal{L}_{\text{sa}}, \quad (8)$$

where λ controls the strength of representation alignment. Setting $\lambda = 0$ recovers the original discrete graph diffusion model, ensuring that self-alignment term in SAGraphDiff acts purely as an auxiliary regularizer without altering the diffusion process.

IV. EXPERIMENTS

We evaluate SAGraphDiff on four molecular graph benchmarks, with ablation studies to identify stable configurations before fixing them for all subsequent experiments.

A. Experimental Setup

a) *Datasets.*: We evaluate on QM9 [14], MOSES [15], GuacaMol [16], and ZINC [17], covering a range of molecular sizes and chemical diversity. We follow the experimental protocols and evaluation metrics and procedures commonly adopted in prior work [4] to ensure consistent and reproducible comparisons across datasets. All models are trained with the AdamW optimizer ($\text{lr} = 2 \times 10^{-4}$, weight decay $= 1 \times 10^{-12}$). The EMA teacher momentum is set to $\alpha = 0.999$, and the projection head is a single linear layer. All reported results are averaged over three independent runs.

B. Ablation Studies

a) *Ablation Studies on QM9.*: We conduct ablation studies on the QM9 dataset. QM9 provides a controlled benchmark with relatively small molecular graphs and shorter training cycles, making it well suited for systematic ablation analysis. In particular, we examine how the scheduling strategy and magnitude of the alignment weight λ , as well as the selection of graph transformer block layers used for representation alignment, affect molecular validity, uniqueness, and novelty.

b) *λ Strategy and Range.*: We first analyze the effect of different alignment weight scheduling strategies. We analyze two alignment weight scheduling strategies. The first, *Decay $\rightarrow$ Constant*, applies strong alignment early and gradually reduces it to a stable scale. The second, *Constant $\rightarrow$ Decay*, maintains fixed alignment initially before decaying to a lower scale late in training.

In addition to the scheduling strategy, we also explore the effect of the overall magnitude of the alignment weight by varying the λ range while keeping the graph transformer block layers fixed. As summarized in Table I, across almost all configurations, SAGraphDiff yields substantial improvements in molecular novelty while largely preserving molecular validity and uniqueness. For example, under the Decay $\rightarrow$ Constant strategy with a λ range of $20 \rightarrow 10$, novelty improves from 40.03 to 47.96 (a relative increase of approximately 20%), while validity and uniqueness remain nearly unchanged ($99.00 \rightarrow 99.24$ and $96.20 \rightarrow 97.03$, respectively).

Overall, the Decay $\rightarrow$ Constant strategy with a λ range of $20 \rightarrow 10$ yields the most favorable trade-off, achieving the largest improvement in novelty without sacrificing validity or uniqueness. When λ is set too large, the alignment loss begins to dominate training and interferes with the diffusion objective, leading to a drop in molecular validity. We therefore adopt this configuration in all subsequent experiments.

c) *Graph Transformer Block Layers.*: Then, we analyze the effect of applying alignment at different graph transformer block layers, as reported in Table I. This design is motivated by prior work showing that deeper layers in transformer networks capture more abstract and semantically

TABLE I
ABLATION STUDY ON QM9.

λ Strategy	λ Range	Block Layers	Valid $\uparrow$	Unique $\uparrow$	Novel $\uparrow$
Baseline			99.00	96.20	40.03
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	3 $\rightarrow$ 7	99.24	97.03	47.96
Constant $\rightarrow$ Decay	20 $\rightarrow$ 10	3 $\rightarrow$ 7	99.21	96.99	46.98
Decay $\rightarrow$ Constant	2 $\rightarrow$ 1	3 $\rightarrow$ 7	99.22	96.96	44.65
Decay $\rightarrow$ Constant	10 $\rightarrow$ 5	3 $\rightarrow$ 7	99.23	96.99	45.61
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	3 $\rightarrow$ 7	99.24	97.03	47.96
Decay $\rightarrow$ Constant	50 $\rightarrow$ 25	3 $\rightarrow$ 7	98.97	96.99	49.19
Decay $\rightarrow$ Constant	100 $\rightarrow$ 50	3 $\rightarrow$ 7	98.63	96.88	50.49
Decay $\rightarrow$ Constant	200 $\rightarrow$ 100	3 $\rightarrow$ 7	99.15	96.67	50.63
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	1 $\rightarrow$ 9	99.34	97.01	46.51
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	2 $\rightarrow$ 8	99.27	96.78	46.80
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	3 $\rightarrow$ 7	99.24	97.03	47.96
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	1 $\rightarrow$ 5	98.91	97.01	46.25
Decay $\rightarrow$ Constant	20 $\rightarrow$ 10	5 $\rightarrow$ 9	98.93	96.94	51.02

TABLE II
MOLECULE GENERATION RESULTS ON MOSES.

Method	Val $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	Filters $\uparrow$	FCD $\downarrow$	SNN $\uparrow$	Scaf $\uparrow$
DiGress [4]	85.7	100	95.0	97.1	1.19	0.52	14.8
SAGraphDiff (ours)	86.8	100	94.9	97.1	1.21	0.52	16.9

stable representations [11], making them suitable as supervisory signals for guiding shallower, weaker representations toward more informative feature spaces. In our experiment, the graph transformer backbone consists of 9 layers in total, which defines the candidate range for selecting student and teacher representations. Accordingly, a block layer range (e.g., 3 $\rightarrow$ 7) indicates that the representation extracted from the 3rd layer is treated as the student, while the representation from the 7th layer serves as the teacher.

Balancing these trade-offs, the 3 $\rightarrow$ 7 configuration offers the most favorable overall behavior, achieving substantial novelty improvements while maintaining strong validity and uniqueness. We therefore adopt this setting for all subsequent experiments.

C. Main Results on Molecular Graph Generation

a) MOSES Molecular Dataset.: Results on the MOSES benchmark are reported in Table II. Following [4], we assess validity, novelty, filter scores, scaffold diversity, and SNN.

SAGraphDiff improves validity (85.7 $\rightarrow$ 86.8) and scaffold diversity (14.8 $\rightarrow$ 16.9), while other metrics remain comparable. Overall, these results suggest that SAGraphDiff promotes more diverse and reliable molecular generation while preserving other core generative behavior of discrete graph diffusion models.

b) GuacaMol Molecular Dataset.: Table III reports results on the GuacaMol benchmark. Some prior methods achieve high scores on individual metrics through hard constraints rather than learned distributions. For instance, near-

perfect validity alongside low KL-divergence. We therefore focus comparisons on discrete diffusion-based models, where all metrics reflect a jointly learned generative process.

Under this setting, incorporating self-alignment into DiGress leads to a clear improvement in molecular validity, increasing from 85.2 to 88.4, while maintaining comparable uniqueness and novelty. KL-divergence is also improved (92.9 $\rightarrow$ 94.4), indicating better distribution matching.

TABLE III
MOLECULE GENERATION RESULTS ON GUACAMOL.

Model	Val $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	KL div $\uparrow$
NAGVAE [18]	92.9	95.5	100	38.4
MCTS [4]	100	100	95.4	82.2
DiGress	85.2	100	99.9	92.9
SAGraphDiff (ours)	88.4	100	99.3	94.4

c) ZINC Molecular Dataset.: Results on the large-scale ZINC benchmarks are summarized in Table IV.

Compared to the DiGress baseline, SAGraphDiff leads to consistent improvements in molecular validity on ZINC (83.5 $\rightarrow$ 87.6). At the same time, near-perfect uniqueness and novelty are preserved.

D. Training Efficiency and Convergence Behavior

Figure 3 shows novelty evolution on QM9. We focus on novelty as validity and uniqueness quickly saturate for both models. As training progresses, novelty decreases for both the baseline and SAGraphDiff, reflecting the common quality-diversity trade-off in molecular generation.

TABLE IV
MOLECULE GENERATION RESULTS ON ZINC.

Model	Valid $\uparrow$	Unique $\uparrow$	Novel $\uparrow$
GraphNVP [19]	42.6	94.8	100
MoFlow [20]	81.7	100	100
GRF [21]	73.4	53.7	100
DiGress	83.5	100	100
SAGraphDiff (ours)	87.6	100	100

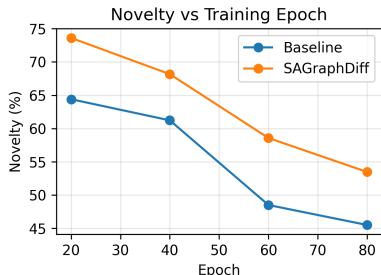


Fig. 3. Novelty versus training epochs on the QM9 dataset.

As shown in Figure 3, SAGraphDiff consistently maintains higher novelty than the baseline across all training epochs, while achieving comparable validity. Table V further quantifies this effect by reporting the number of training epochs required for convergence on multiple molecular benchmarks. Across most datasets, incorporating self-alignment tends to reduce the required training epochs, suggesting that representation alignment may contribute to improved training efficiency alongside generation performance. The gains are more pronounced on GuacaMol and MOSES, suggesting greater efficiency benefits on more complex molecular benchmarks.

TABLE V
NUMBER OF TRAINING EPOCHS FOR (I) THE BASELINE DIGRESS MODEL TO REACH THE REPORTED PERFORMANCE, AND (II) THE SAGRAPHDIFF MODEL TO REACH THE REPORTED PERFORMANCE, ACROSS DIFFERENT DATASETS.

Dataset	Baseline Epochs	SAGraphDiff Epochs
QM9	100	80
ZINC	165	140
GuacaMol	1000	60
MOSES	300	230

V. CONCLUSION

We propose SAGraphDiff, which incorporates self-alignment into discrete graph diffusion for one-shot molecular graph generation. Experiments show consistent improvements in validity and diversity, suggesting self-alignment as a general and lightweight mechanism for improving discrete graph diffusion models.

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